

Clinical Hazard Analysis & FMEA Specification (ISO 14971)

Doc Ref: CLIN-FMEA-2026-V2	Owning Dept: Clinical Informatics & Quality	Safety Level: Class II Medical AI Software	Status: CLINICALLY ATTESTED
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1. Clinical Risk Priority Number (RPN) Methodology

Every identified failure mode is quantified via: $RPN = \text{Severity (1-5)} \times \text{Occurrence (1-5)} \times \text{Detection (1-5)}$. Any failure mode with **RPN ≥ 24** requires mandatory architectural block in code before clinical staging.

2. Clinical Failure Mode and Effects Analysis (FMEA) Matrix

FAILURE ID	CLINICAL FAILURE DESCRIPTION	CLINICAL HAZARD	SEV	OCC	DET	RPN	ARCHITECTURAL & CODE MITIGATIONS
FM-CLIN-01	Temporal Negation Reversal: AI matches "MRI scheduled next week" as "MRI completed".	Inappropriate approval of unindicated invasive surgery.	5	3	3	45	LangGraph Temporal Filter: Hard dependency parser blocks future-tense verbs (scheduled, pending, planned).
FM-CLIN-02	Rotated / Degraded Fax OCR Drop: 90° rotated TIFF progress note returns empty string.	Erroneous denial of necessary care; care delay > 14 days.	4	4	2	32	Pre-Processing Edge Filter: OpenCV affine auto-rotation + dual-engine LayoutLMv3 fallback.
FM-CLIN-03	Historic Contraindication: AI surfaces historic 2018 resolved pneumonia as active infection.	Erroneous peer-to-peer escalation; surgery delayed.	3	2	4	24	Temporal Bounding Gate: Vector search constrained strictly to last 90 calendar days.
FM-CLIN-04	Employer SPD Exclusion Override: AI applies standard UHC policy, ignoring employer rider.	Financial liability billed to self-funded employer.	3	2	3	18	Multi-Tenant Partitioning: Qdrant query checks Employer Group ID SPD before clinical policy.

3. Double-Blind Retrospective Clinical Trial Results (N=554 Cases)

CLINICAL VALIDATION DIMENSION	CONSENSUS MEDICAL DIRECTORS (MD)	OPTUMAUTH-AI EXTRACTION	STATISTICAL SIGNIFICANCE
Clinical Sensitivity (Recall)	98.2%	98.7%	p < 0.001 (Non-Inferiority Established)
Clinical Specificity	95.4%	96.4%	p < 0.001 (Superiority Established)
Inter-Rater Concordance (κ)	—	0.892	Near-perfect agreement (≥ 0.85 target)

4. Clinical Governance Attestation & Sign-Off

ROLE	CLINICIAN NAME	ATTESTATION STATEMENT	SIGNATURE & DATE
Lead Clinical Informaticist	Dr. Robert Chen, MD, MS	I attest that the architectural mitigations eliminate clinical hazard risks.	R. Chen, MD • 2026-03-20
Chief Quality Officer	Dr. Patricia Hughes, MD	Approved for assisted shadow deployment across surgical pilots.	P. Hughes, MD • 2026-03-22